Dollatham Charoenthammakit

AI Engineer · AI Cybersecurity · Data Scientist · Data Engineer

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A dedicated Computer Science student at KMUTT with a strong focus on Artificial Intelligence, Machine Learning, Cybersecurity, and Full-Stack Development. Experienced in developing AI-powered systems ranging from eye disease detection to financial market prediction. Passionate about bridging cutting-edge AI research with practical, production-ready applications.

EDUCATION

King Mongkut's University of Technology Thonburi (KMUTT) 2022 — Present

Bachelor of Science in Applied Computer Science · GPAX 3.18

Bangkok, Thailand

WORK EXPERIENCE

AI Cybersecurity Intern 2025

Datafarm Co., Ltd.

Gained hands-on experience applying AI and machine learning to cybersecurity operations. Developed AI-driven threat detection pipelines, automated vulnerability scanning with intelligent triage, and built models for security log anomaly detection. Participated in penetration testing, digital forensics, and leveraged LLM-based tools for accelerated OSINT and incident response.

AI Threat Detection | ML Security Analytics | Penetration Testing | Digital Forensics | LLM for Security | OSINT

AI Trading Intern 2024

Pi Securities Public Company Limited

Applied machine learning, deep learning, and data science techniques to develop intelligent trading systems, analyze financial market data, and optimize trading strategies.

AI Trading | Machine Learning | Financial Markets | Quantitative Analysis

Part-time Developer 2023

Research Center of KMUTT

Worked on a research project on vehicle accidents utilizing AI technology to calculate and assess vehicle damage. Developed the cross-platform application using Flutter for both iOS and Android, ensuring seamless integration with the AI processing system.

Flutter | Dart | iOS | Android | AI Integration | REST API

TECHNICAL SKILLS

Programming Languages: Python, SQL, C++, Java, JavaScript, TypeScript, Dart, Swift

AI / Machine Learning:

PyTorch, TensorFlow, Hugging Face, OpenCV, LSTM/CNN, Transformers, RAG, LLM, LangChain

Data Science & Eng.: Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn, Apache Spark, ETL

Cybersecurity:

Wireshark, Nmap, Burp Suite, Metasploit, SQLMap, Nikto, Hashcat, Aircrack-ng, OSINT, Sherlock, Tor, Ghidra

Frontend: React, Next.js, Tailwind CSS, TypeScript

Backend: Node.js, Express, Django, Flask, FastAPI, NestJS, REST API

Mobile: Flutter, React Native, Dart, Swift, iOS, Android

Database: PostgreSQL, MySQL, MongoDB, Redis, SQLite, Firebase, Supabase

DevOps & Tools: Git, Docker, Kubernetes, GitHub Actions, CI/CD

Operating Systems: Linux, Ubuntu, Kali Linux, Windows, macOS

ACADEMIC PROJECTS

TrendReversal-AI — Market Trend Reversal System

Capstone Project

End-to-end machine learning system predicting financial market trend reversals across 5 global markets (US, UK, Thai, Gold, Bitcoin). Features deep learning models (LSTM, CNN, Transformer), regime-switching strategy via HMM/GMM, multi-objective parameter optimization (NSGA-II), hybrid LLM agent with RAG, and an interactive dashboard.

Python | TensorFlow | LSTM | Transformers | HMM/GMM | NSGA-II | LLM | RAG

GitHub OSINT & Secret-Scanning Engine

Cybersecurity / AI

High-throughput OSINT engine scanning public GitHub repositories at 200 repos/min to automatically detect hard-coded secrets. Features on-premise LLaMA-based AI classification, async networking, Redis caching, ChromaDB vector indexing, and real-time metrics.

Python | LLaMA | Redis | ChromaDB | Docker | Asyncio | OSINT

MongTa — Eye Disease Detection

Software Eng.

Application detecting cataracts, styes, conjunctivitis, and pterygium. Developed multiple models (YOLOv11, Transformers) and set up a server for data transmission to the frontend.

YOLOv11 | Transformers | Roboflow | Computer Vision

SAR to Multispectral Imagery — GANs & Vision Transformers

DS & DE

Satellite image enhancement and colorization using Pix2Pix, SRGAN, and SwinUNet with comprehensive data engineering.

Pix2Pix | SRGAN | SwinUNet | GANs | Vision Transformers

Automated Trading Bot — ML for Crypto Prediction
Capstone Project

Trading bots using machine learning to predict Bitcoin trends and execute real-time trades, integrated with Binance. Used Decision Tree and Hidden Markov Model with feature selection via Gini Index and Entropy.

Decision Tree | HMM | Binance API | Python

Penetration Testing & Vulnerability Assessment
Cybersecurity

Comprehensive penetration testing on servers and websites. Identified vulnerabilities and generated risk assessment reports following international standards.

Nmap | Metasploit | SQLMap | Nikto

Wongyos CTF — Ethical Hacking Challenges
CTF Competition

Solved cybersecurity challenges in Reverse Engineering, Cryptography, Digital Forensics, Web Exploitation, and System Exploitation.

University Network Malware Analysis & Forensics
Malware Analysis

Detected and analyzed backdoor malware using IDA Pro, Ghidra, and HxD to extract hardcoded credentials and uncover unauthorized authentication mechanisms.

Network Traffic Analysis & OSINT Dark Web Investigation
Digital Forensics

Network forensics via Wireshark and OSINT tracking across social platforms and Tor Network (.onion domains).

Melody Music — Music Streaming Platform
Full-Stack

Music streaming website using TypeScript, Next.js, Tailwind CSS, Supabase, deployed with Vercel.

Biz NA — Real-Time Asset Calculator
OOP

Web application that tracks asset portfolios in real-time — cryptocurrencies, stocks, cash, and other volatile assets.

Chefbot — Voice-Interactive Menu Assistant
Programming

Voice-interactive program providing personalized menu suggestions through speech recognition and intelligent recommendation logic.

RELEVANT COURSEWORK

AI & Data
- Artificial Intelligence and Machine Learning
- Natural Language Processing and Generation
- Computational Intelligence
- Data Science and Data Engineering
- Big Data Analytics

Software & Web
- Computer Programming
- Object-Oriented Programming
- Web Programming
- Software Engineering

Systems & Networks
- Operating Systems
- Computer Architecture and Organization
- Network Computing
- Blockchain Technology

Math & Foundations
- Discrete Mathematics
- Linear Algebra for Computing
- Probability and Statistics
- Data Structures and Algorithms
- Database Systems

COMPETITIONS & AWARDS

Winner — Innovation for KMUTT Sustainability 2023
Playground 2023

Honorable Mention — Research to Market 2023
Big's SEED Talent Camp 2023

Regional Qualifiers 2023
Startup Thailand League 2023

CAREER INTERESTS

🤖 AI Engineer
Deep learning, LLM/RAG agents, MLOps, model deployment

🛡️ AI Cybersecurity Engineer
AI threat detection, SIEM & SOAR, malware analysis, penetration testing

📊 Data Scientist
Statistical modeling, feature engineering, predictive analytics

⚙️ Data Engineer
ETL pipelines, data warehousing, Apache Spark, cloud infrastructure